UNIT V - ADVANCED MATERIALS

1. **Distinguish between active and passive display units?**
   Active display device: These devices display or emit the light radiation in their own due to application of field.
   Example: LED – Light Emitting Diode
   Passive display device: These devices modulate the incident radiation and then collect or transmit the light.
   Example: LCD- Liquid Crystal Display

2. **List out the differences between LED and LCD.**

Sl.No	LED	LCD
1	Cost is high.	Cost is very low.
2	Not suitable for large area display.	Suitable for large area display.
3.	High consumption power (Milliwatts).	Low power consumption (microwatts)
4.	Response time is in nano seconds (10-9 seconds).	Response time is in microseconds (10-6 seconds).
5	Intensity of light can be controlled	Intensity of light cannot be controlled
6	Different colur displays are available at low cost.	Colour displays will not be available at low cost.
7	Operating temperature is 0^o to 70^oC	Operating temperature is 10^oC to 47^oC

3. **What is meant by LCD? Give its principles and types**
   LCD is the abbreviation of Liquid Crystal Display. It is a digital display unit in which the digits are displayed by the principle of transmission or reflection of the incident beam.
   Types: (i) Dynamic scattering display
   (ii) Twisted nematic display.

4. **What is a liquid crystal? Give its various phases**.
   Liquid crystals are organic fluids in cigar shape and have the phases between liquid and a crystal. The various phases of liquid crystals are
   (i) Smetic phase (ii) Nematic phase

5. **What is meant by MBBA molecule? Give its structure.**
   4-Methoxy –Benzylidene, 4- Butylaniline is called as MBBA molecules. It is formed by two benzene rings linked in a single group.

6. **Define Director and Pitch.**
   Director: In liquid crystals the molecules will have a preferred orientation or direction is called director
   Pitch: The distance between two similar plane in which the molecule has same director is called pitch.

7. **Out of LED and LCD which is more advantages? Why?**
   Out of LED and LCD, LCD is more advantages because of the followimg reasons
   (i) It has low cost
   (ii) It has low power consumption
   (iii) It occupies smaller area
   (iv) It has many applications in all digital equipments.

8. **Give any four applications of LCD? (Or)What are the applications of display devices?**
   (i) They are used in wrist-watches, clocks
   (ii) They are used in calculators, measuring instruments like digital multimeter
   (iii) Digital balance also has application of display devices
   (iv) They are used in measuring the temperature (Digital Thermometers)
   (v) They are used in Digital Gauss Meters.

9. **What are Metallic glasses?**
   Metallic glasses are the amorphous metallic solids which have high strength, good magnetic properties and better corrosion resistance and will possess both the properties of metals and glasses.

10. **What is meant by glass transition temperature?**
   The temperature at which the metals (alloys) in the molten form transform into glasses i.e. liquids to solids is known as glass transition temperature (Tg).

11. **What do you understand by the term Quenching?**
   'Quenching' is a technique used to form metallic glasses, quenching means rapid cooling. Actually atoms of any materials move freely in a liquid state Atoms can be arranged regularly when a liquid is cooled slowly. Instead, when a liquid is quenched, there will be an irregular pattern, which results in the formation of metallic glasses.

12. **What are the types of metglasses?**
   Metallic glasses are of two types
   (i) Metal-metalloid glasses
   Examples: Metals- Fe, Co, Ni
   Metalloids – Ge, Si, B,C
   (ii) Metal-Metal glasses
   Examples: Metals-Ni, Mg, Cu
   Metals – Niobium, Zn, Zr

13. **Mention any four properties of metglasses?**
   (i) Metallic glasses have tetrahedral closely packed (TCP) structure rather than hexagonal closely packed (HCP) structure.
   (ii) The Metallic glasses are very strong in nature
   (iii) They possess malleability, ductility etc
   (iv) They exhibit very low hysteresis loss and hence transformer core loss is very less.

14. Write any four applications of metglasses.

1. Since the metallic glasses possess low magnetic loss, high permeability, saturation magnetization and low coercivity, these materials are used in cores of high power transformers.
2. Since the metallic glasses are very strong and hard they are used to make different kinds of springs.
3. Because of their high resistivity, they are used to make computer memories, magneto-resistance sensors etc.
4. Since the metglasses are not affected by irradiation, they are used in nuclear reactors.
5. Dielectric strength: When external field applied to a dielectric material is greater than the critical field, the dielectric losses its insulating property and becomes conducting. Therefore a large current flow through the material. This phenomenon is called dielectric breakdown.

15. What are nano-phase materials? Give examples

Nano-phase materials are the materials in which the grain size is in the order of 1to100 nano-meters and these atoms will not move away from each other. They exhibit greatly altered Physical, Chemical and Mechanical properties compared to their normal, large grained counterparts with the same chemical composition.
Examples: Zno, Cu-Fe alloys, Ni, Pd, Pt etc.

16. **Give any four techniques adopted to synthesis the nano-phase materials**

(i) Mechanical crushing or ball milling
(ii)Chemical vapour deposition
(iii)Electro-deposition
(iv)Laser synthesis

17. **Classify Nanomaterials based on their grain size.**

Sl.No	Types	Crystallite / grainsize / shape
1.	Zero dimension	Clusters or powders (quantum dots)
2	One dimension	Multilayer
3	Two dimension	Ultra fine grained over or buried layers
4	Three dimension	Equiaxed nanometer sized grains

18. What are the two ways adopted to synthesis nano particles?

The two ways of synthesizing nano particles are
(i) Top-Down approach
(ii) Bottom-Up approach

19. **What are different types techniques used in Top-Down and Bottom-Up approach for synthesizing nano materials?**

Top-Down approach:

1. Mechanical Alloying
2. Lithography
3. Errosion etc.,

Bottom-Up approach:

1. Chemical precipitation
2. Gas Phase agglomeration
3. Self assembly etc.,

20. What are the important properties of nano materials?

(i) The size of the nano-material will be in the order of 1-100nm and hence the inter particle spacing is found to be very less and they do not have any dislocation in it.

(ii) Nano-materials have high strength and super hardness

(iii) The energy bands in these materials will be very narrow and the ionization potential is found to be higher

(iii) Nano-materials posses local magnetic moment within themselves, and these materials exhibits spontaneous magnetization at smaller sizes

21. List out any four recent applications of nano- materials

i) Nano-MEMS (Micro-Electro Mechanical Systems) are used in ICs, Optical Switches, Pressure sensors, mass sensors etc.,

(ii) Recently nano-robots were designed, which are used to remove the damaged cancer cells and also to modify the neuron network in human body.

(iii) Nano-dimensional photonic crystals and quantum electronic devices plays a vital role in the recently developed computers.

(iv) Bio-sensitive nano-particles are used in the production of DNA-chips, bio-sensors etc.,

22. **What is the difference between Nano materials and Macro materials?**

All materials are composed of grains, which in turn comprise many atoms. Depending on the size these grains may be visible or invisible to the naked eye. Conventional materials or Macro materials have grain of size varying from hundreds of microns to centimeters. Nanomaterials have grain size in the order 1-100nm. Such grains contain maximum of few hundreds of atoms.

23. What are Carbon Nano tubes or CNT?

Carbon nano tubes are sheet tubes formed by rolling the graphite (or graphene) sheet into tubes with the bonds at the end of the sheet. These bonds are used to close the tube. They posses exotic properties like very low specific resistivities in metallic carbon nano tubes and high mobilities for semi conducting nanotubes.

24. List out the types of CNT

The two types of Carbon nano tubes are

1. Multi walled carbon nanotubes (MWCNT)
2. Single walled carbon nanotubes (SWCNT)

25. List out the structures of CNT.

Carbon nano tubes are having three different structures depending upon the orientation of C-C bonds. They are

1. Armchair structure
2. Zig Zag structure
3. Chiral Structure

26. **Mention any some important applications of Carbon nano tubes**.

(i) The nanotubes are used for electron emission effect in vacuum lamps. The resultant lights are much brighter than the conventional light.

(ii) Nano metal wires with small diameters are used to do the interconnections between the computers and switches

(iii) Carbon nano tubes are used as a computer switching device

(iv) Carbon nanotubes have been used as a storage device in battery, fuel cells, etc.,

(v) Chemical sensors are made using CNT to detect various gases like NO2.

(vi) CNT is used as a catalyst for enhancing the chemical reactions.

27. What are shape memory alloys?

When material is heated above the transformation temperature, then there will be some change in the crystal structure, which cause the material to return to its original structure. Such materials are called shape memory alloys.

28. What is transformation temperature?

In SMA, the shape recovery process occurs not at a single temperature rather it occurs over a range of temperature.

Thus, the range of temperature at which the SMA switches from new shape to its original shape is called transformation temperature or memory transfer temperature.

29. What do you understand by martensite and austenite phases?

Martensite is an interstitial super solution of carbon in α-iron and it crystallizes into twinned structure. The SMA will have this structure generally at lower temperature and it is soft in this phase.

Austenite is the solid solution of carbon and other alloying elements in γ –iron and it crystallizes into cubic structure. The SMA will attain this structure at higher temperatures and it is hard in this phase.

30. What is meant by shape memory effect?

At lower temperature the SMA will be in martensite structure and when it is heated then it will change its shape toaustenite structure and while cooling it will again return to martensite form. This effect is called shape memory effect.

31.Define Pseudo –elasticity?

Pseudo –elasticity occur in some types of SMA in which the change in its shape will occur even without in its temperature.

32. What are the properties of SMA?

1. The transformation occurs not only at a single temperaturerather they occur over a range of temperature.
2. They have Pseudo –elasticity and super elastic property.
3. They exhibit hysteresis curve, during cooling and heating process.
4. Crystallographically the thermo –elasticmartensites are reversible.

33. What are the types of SMA?

1. One way shape memory alloy (SMA):

Though there is some change in its temperature, the SMA remains in the same phase, and this type of material is called one way shape memory alloy.

2. Two way SMA: The type of materials which produces spontaneous and reversible deformation just upon heating and cooling even without load are called two way shape memory alloys.

34. Give any 4 applications of SMA?

1. They are used in relays and activators
2. SMA is used in eye glass frames , toys , helicopter blades , blood clot filter etc.
3. They are also used in fire safety valves, coffee maker, circuit edge connector etc.
4. Ni-Ti SMA are also used in artificial hip-jionts. Bone-plates pins for healing bones –fractures and also in connecting broken bones.